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DEPARTMENT OF MECHANICAL ENGINEERING

I / II Semester B.E. Degree

Elements of Mechanical Engineering [22ME110/22ME210]

Faculty Name: Karthik MS

QUESTION BANK

UNIT - 5

Refrigeration:

- (1) Define the following: (i) **Refrigeration** (ii) **Ton of Refrigeration**, (iii) **COP**, (iv) **Refrigerating effect** (v) **Refrigerant**, (vi) **Ice making capacity** (vii) **Relative COP**.
- (2) List the **application of refrigeration**.
- (3) Explain with a neat sketch, **Vapor compression refrigeration (VCR) system**.
- (4) Explain with a neat sketch, **Vapor absorption refrigeration (VAR) system**.
- (5) **Distinguish** between **VCR** and **VAR system**.
- (6) Define refrigerants, **types of refrigerants** and List the **desirable properties of refrigerants**.

Air Conditioning:

- (1) What do you mean by **air conditioning** and list some **application of air conditioning**.
- (2) Explain with a neat sketch, **window type room air conditioner**.

Robotics:

- (1) Define **Robots** and **classification of robots**.
- (2) Explain the following with a **neat sketch**: (i) **Cartesian coordinate robot** (ii) **Cylindrical coordinate robot** (iii) **Spherical coordinate robot**.
- (3) List the **application, advantages and disadvantages of robots**.

Automation:

- (1) Define **automation** and Explain the **types of automation**:(i) **Fixed automation** (ii) **Programmable automation** (iii) **Flexible automation**.
- (2) List the **advantages** and **disadvantages** of **automation**.